PROJECT EXPO 2026

Smart Hospital Care System

AI-Based Patient Monitoring and Predictive Alert System

TEAM MEMBERS:

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Smart Hospital Care System is an embedded healthcare monitoring solution designed to reduce manual monitoring workload and support faster nursing response. The system focuses on two key parameters: **IV fluid status** and **patient bed occupancy**. A load cell with HX711 is used to monitor the IV fluid weight and estimate the remaining fluid and time, while multiple FSR sensors detect patient presence and bed-vacancy events.

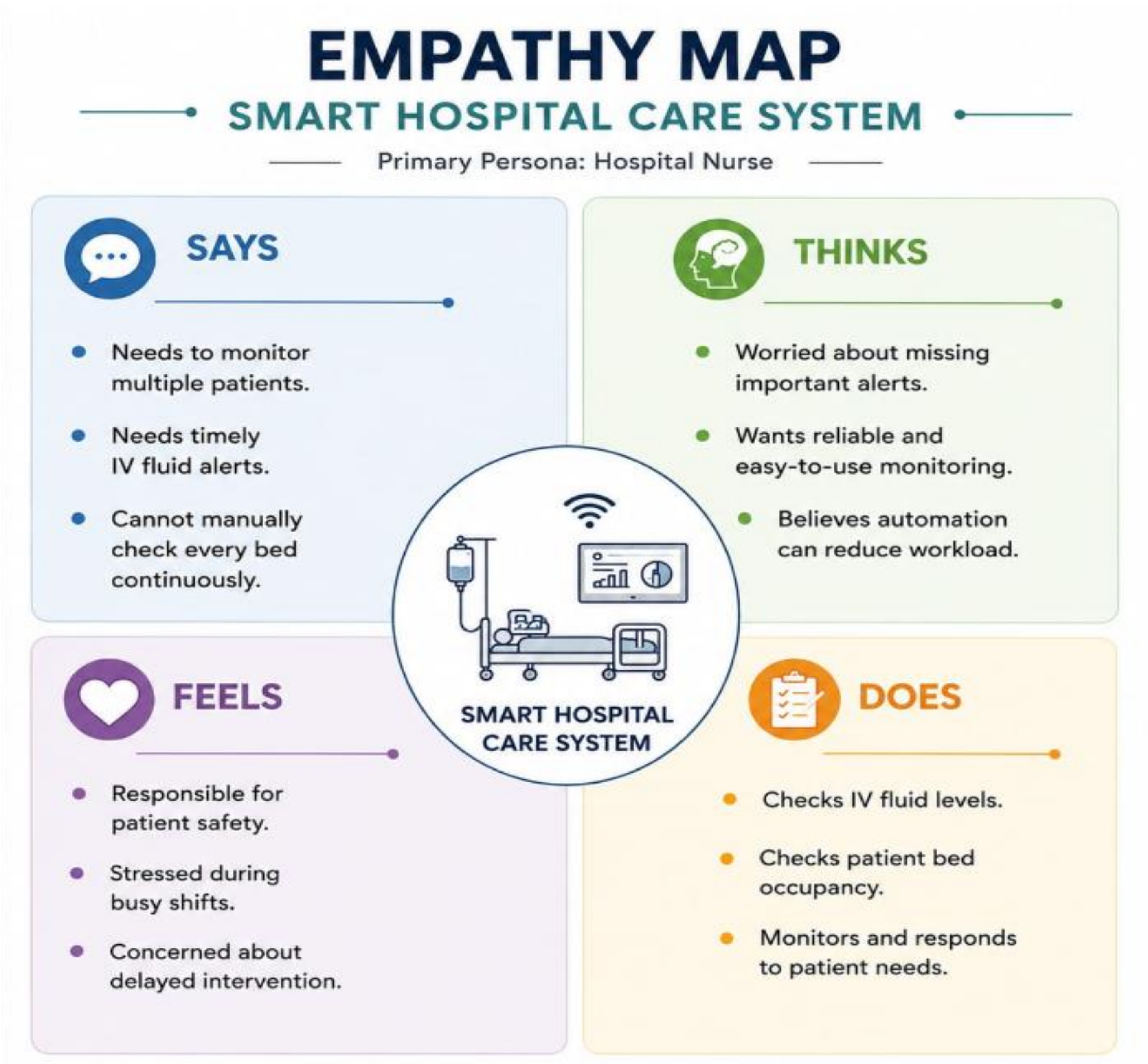
Brainstorming Report:

Idea Prioritization Matrix:

Idea	Feasibility	Innovation	Cost	User Benefit	Priority
IV Fluid Monitoring	High	High	Low	Very High	Very High
Bed Occupancy Monitoring	High	Medium	Low	High	High
Predictive IV Time	Medium	Very High	Low	Very High	Very High
Wireless Dashboard	High	High	Medium	High	High
Patient Attention Score	Medium	High	Low	High	High

Key Findings:

- IV fluid monitoring was selected as the primary feature because it directly improves patient safety.
- Bed occupancy monitoring enhances patient supervision with minimal hardware.
- Predictive IV time estimation provides proactive alerts instead of reactive notifications.
- Wireless monitoring reduces the need for frequent manual inspection.



User Persona:

Name	: Priya Sharma
Age	: 32 Years
Occupation	: Staff Nurse
Experience	: 6 Years

Responsibilities

- Monitor IV fluid levels.
- Observe patient bed occupancy.

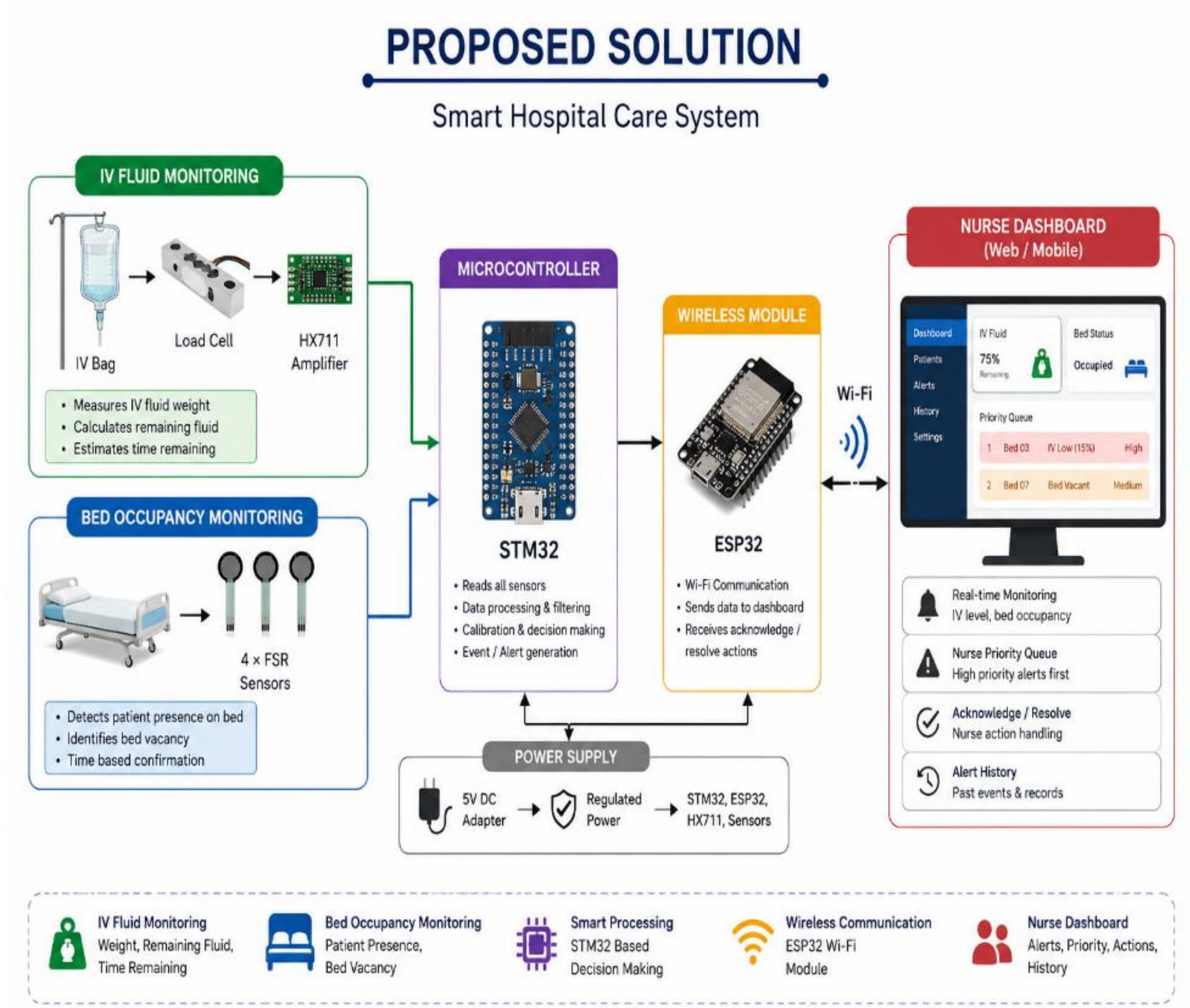
- Respond to patient emergencies.
- Maintain nursing records.

Goals

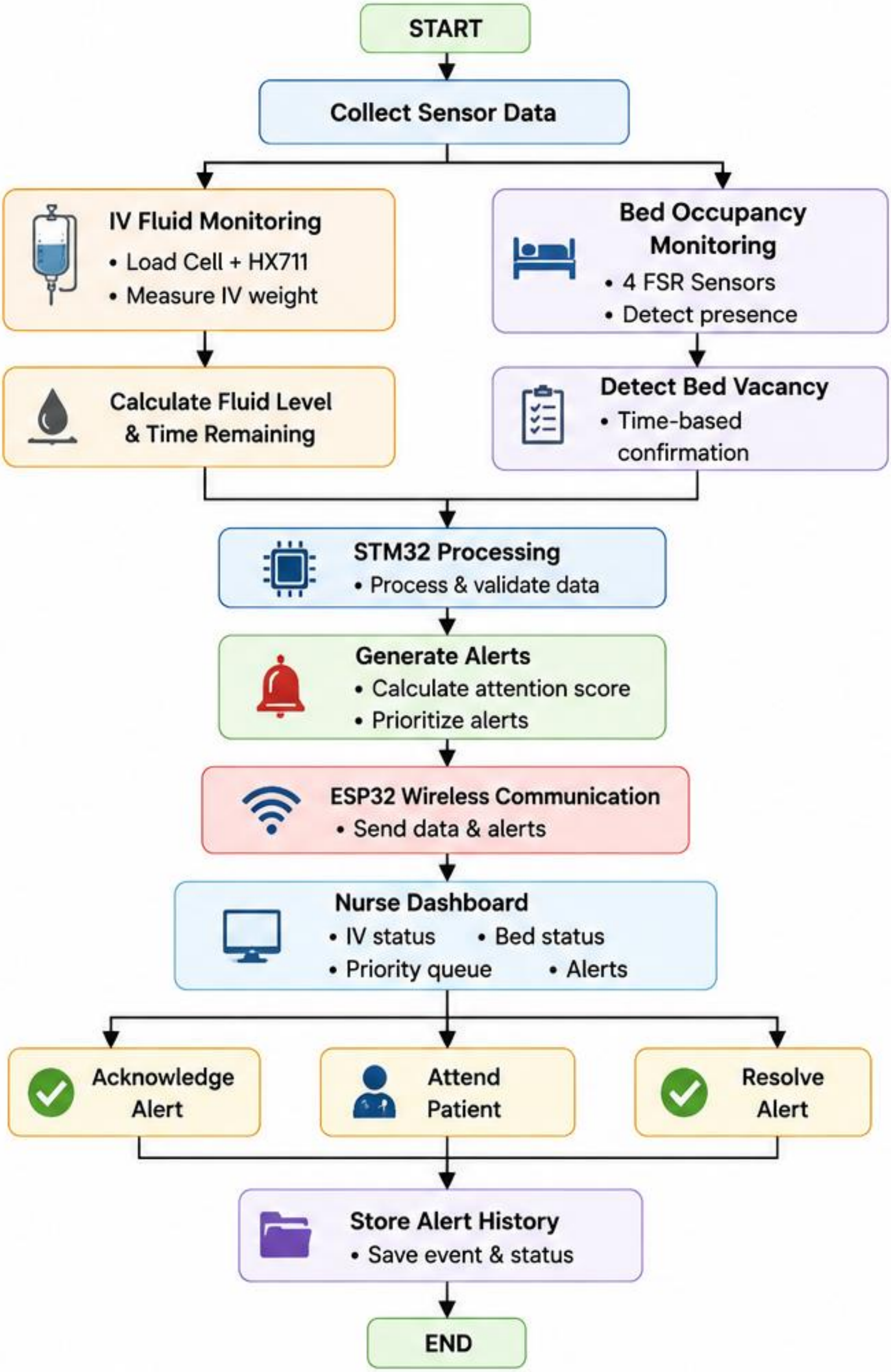
- Reduce manual monitoring.
- Improve patient safety.
- Receive timely alerts.
- Save time during rounds.

Challenges

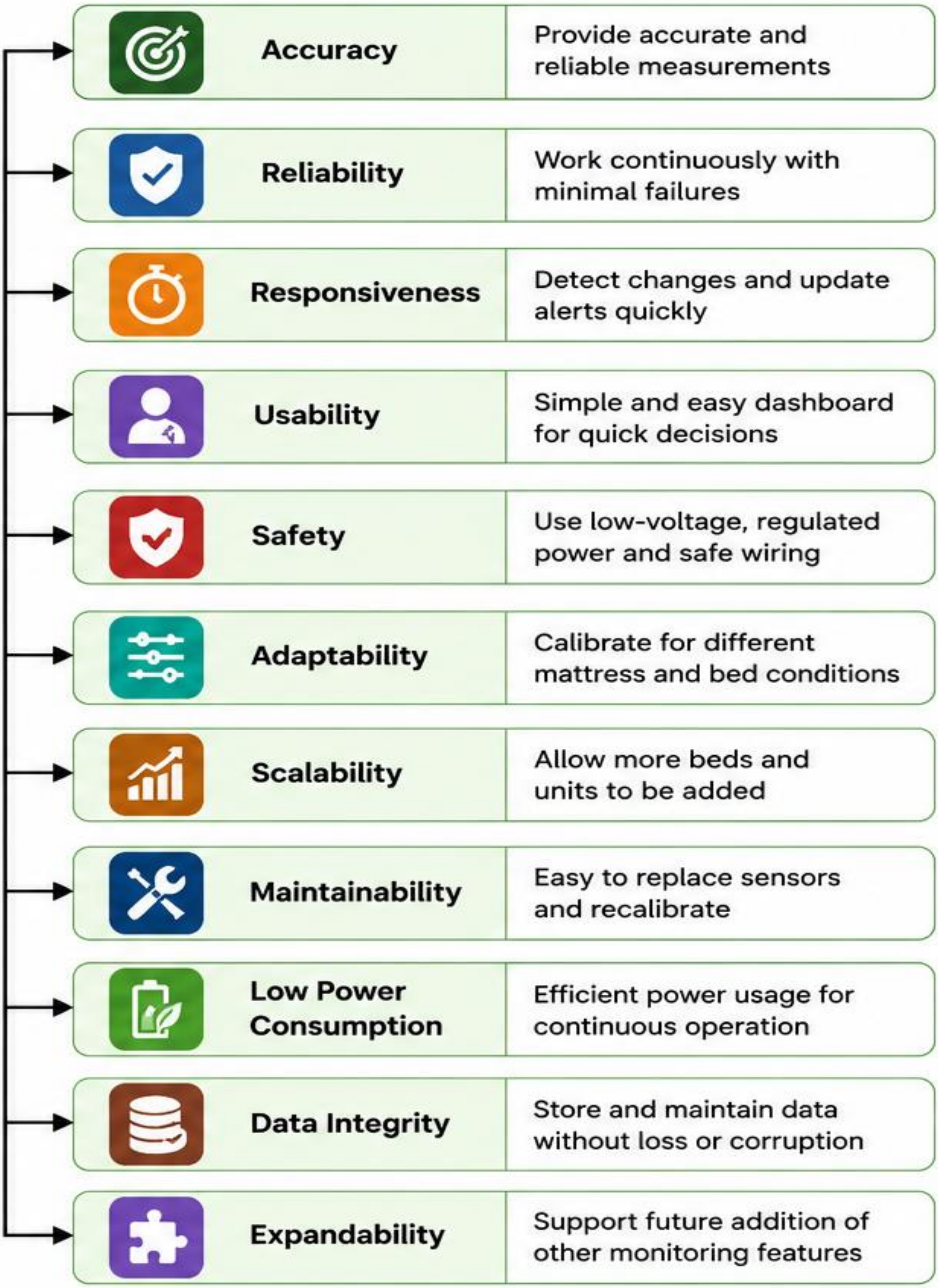
- Monitoring multiple patients simultaneously.
- Delayed identification of empty IV bottles.
- Increased workload during busy shifts.



Functional Requirements:



Non-Functional Requirements:



Project Planning:

Activities	Aug W1	Aug W2	Aug W3	Aug W4	Sep W1	Sep W2
Requirement Analysis	■	■				
Component Procurement	■	■				
Hardware Assembly		■	■			
Sensor Interfacing			■	■		
STM32 Programming			■	■		
ESP32 Communication				■	■	
Dashboard Development				■	■	
Predictive IV Time					■	
Patient Attention Score					■	
System Integration					■	■
Testing & Calibration					■	■
Documentation						■

Legend:

■ =Planned Activity

RISK ANALYSIS

ID	RISK	POTENTIAL IMPACT	LIKELIHOOD	MITIGATION STRATEGY	CONTINGENCY PLAN
R1	**Sensor Measurement Inaccuracy** May affect fluid level and alerts.	Slight error in measurement may lead to wrong estimation.	Medium	• Proper calibration • Use quality sensors • Regular calibration check	• Recalibrate the system • Replace sensor if needed • Use safe threshold limits
R2	**Wireless Communication Issue** May cause delay in alerts.	Alerts or data may be delayed temporarily.	Medium	• Stable Wi-Fi connection • Check signal strength • Auto-reconnect in code	• Store data locally • Reconnect automatically • Alert when connection lost
R3	**System/Software Failure** System may stop unexpectedly.	Monitoring may stop and some alerts may be missed.	Low	• Watchdog timer • Stable code & testing • Regular system reset	• Auto restart using watchdog • Manual reset option • Check logs and restart
R4	**Power Failure** System may turn off.	Monitoring may stop during power failure.	Medium	• Regulated power supply • Use UPS / Power bank • Brownout protection	• Use backup power • Save data locally • Restore system safely
R5	**User Error** Wrong operation or settings.	May lead to wrong actions or delayed response.	Low	• Simple and clear UI • Proper labeling • Basic user training	• Provide confirmation prompts • Add help / guidance • Allow correction of actions

LIKELIHOOD LEVEL

High – Likely to occur | Medium – May occur | Low – Unlikely to occur

Impact: Effect of risk on system performance or project.
Mitigation: Action taken to reduce or prevent the risk.
Contingency: Action taken if the risk occurs.